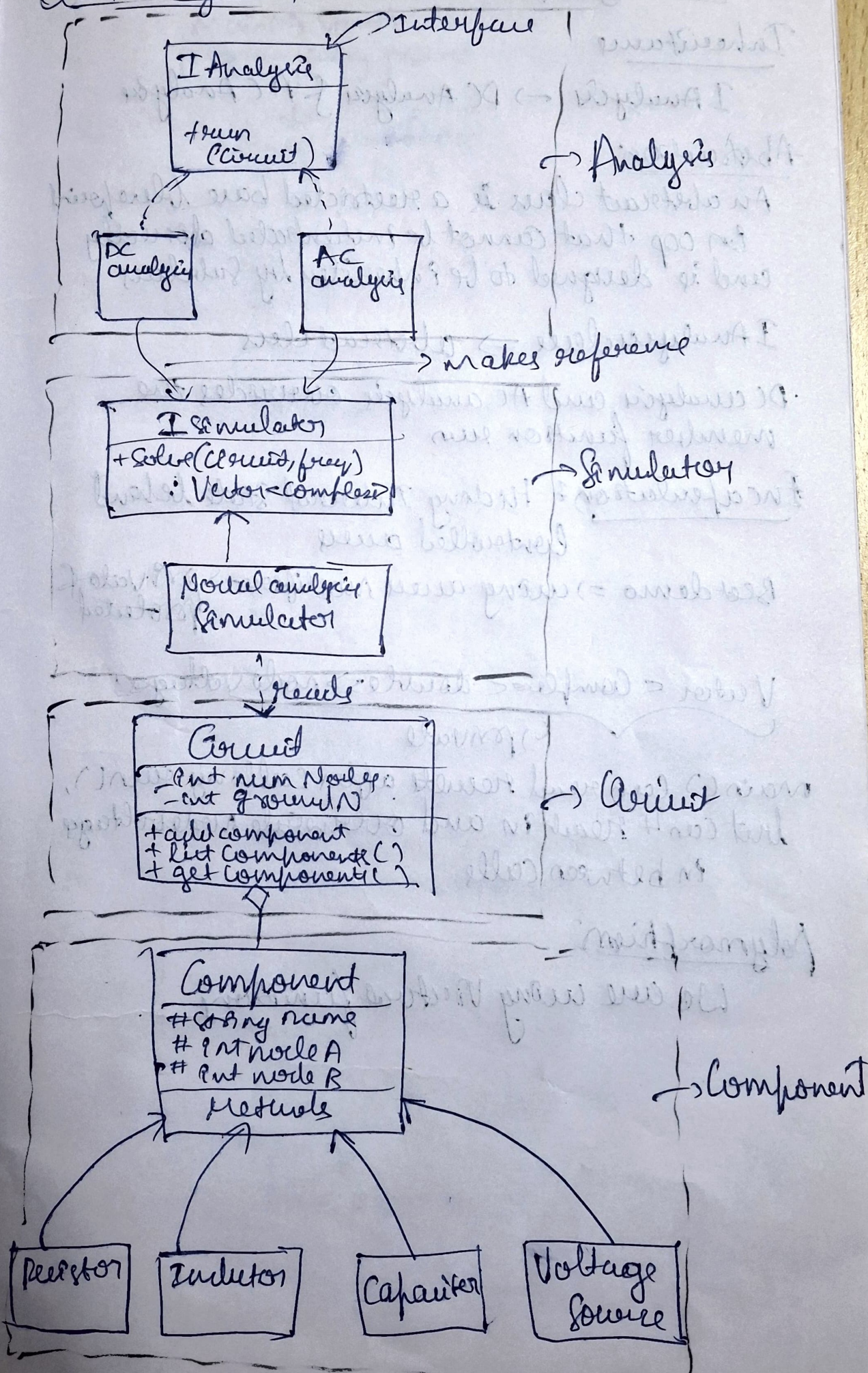


Class diagram



Mainfile Workflow

main() starts
reads node count and creates
circuit (numNodes)

↓
Circuit builder
adds Voltage sources and passive
components

↓
Finish Circuit entry
list components, then runAnalysis(circuit)

↓
Any Voltage Source
has frequency > 0?

No

↓
DC Analysis
solve once at 0 Hz and
print node voltages

Yes

↓
main() requests Sweep
start freq, end freq and
step

↓
ACAnalysis::run(circuit)
clear results and
iterate over sweep
frequencies

for each frequency f in Sweep

MNASimulator::Solve(circuit, f)

↓
SolveLinear(A, b)

↓
Store AC Result(f , node voltage)

↓
Main() prints AC results
node-voltage magnitude and
phase at each frequency